

Sums of powers of integers with a fixed number of binary transitions, and the moments of striped random generators

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Abstract

For $0 \leq k \leq n-1$, let $T_{n,k}$ be the set of integers in $[0, 2^n)$ whose n -bit binary representations have exactly k transitions (adjacent unequal bits), and let $S_{n,k,p} = \sum_{m \in T_{n,k}} m^p$. We show that the generating polynomial of the p -th central power sums over k satisfies a *triangular* linear recursion in n , and therefore admits an explicit eigen-expansion: a closed form as a finite combination of n -th powers of the eigenvalues $2^j(1 + (-1)^j t)$ with explicit rational coefficients in the transition-marking variable t . Extracting the coefficient of t^k yields exact formulas for $S_{n,k,p}$ with $O(k+p)$ terms, and the recursion itself evaluates all k simultaneously in $O(np^2)$ big-integer operations. We give fully explicit results for $p \leq 4$, correct a previously conjectured special case, and derive the exact variance and kurtosis—both for the uniform distribution on $T_{n,k}$ and for the “striped” random generators that motivated the question, in which each bit repeats the previous one with a fixed probability. In the striped model the formulas collapse to short rational expressions; for mean run length m , writing $\beta = 1 - 2/m$, the kurtosis converges as $n \rightarrow \infty$ to $\frac{3(2-\beta)(6+\beta)(8+7\beta)}{5(8-\beta)(2+\beta)^2}$. All formulas have been verified in exact arithmetic against exhaustive enumeration.

1 Setup and notation

Fix $n \geq 1$. Every $m \in [0, 2^n)$ has bits $b_{n-1} \cdots b_0$, and its number of transitions is $\tau(m) = \#\{j \in [1, n-1] : b_j \neq b_{j-1}\}$. Let $T_{n,k} = \{m : \tau(m) = k\}$ and

$$S_{n,k,p} = \sum_{m \in T_{n,k}} m^p.$$

Counting is classical: a string with k transitions is determined by its transition set (a k -subset of the $n-1$ boundaries) and its bottom bit, so $|T_{n,k}| = 2 \binom{n-1}{k}$.

It is convenient to center. Write each bit as a sign $\epsilon_i = 2b_i - 1 \in \{\pm 1\}$ and set

$$c(m) = 2m - (2^n - 1) = \sum_{i=0}^{n-1} \epsilon_i 2^i.$$

Factor out the bottom sign: $\epsilon_i = \epsilon_0 \prod_{j \leq i} s_j$, where $s_j = \epsilon_j \epsilon_{j-1} \in \{\pm 1\}$ equals -1 exactly at transitions. The map $m \leftrightarrow (\epsilon_0, s_1, \dots, s_{n-1})$ is a bijection, $\tau(m) = \#\{j : s_j = -1\}$ does not depend on ϵ_0 , and

$$c(m) = \epsilon_0 Y, \quad Y = Y(s) = \sum_{i=0}^{n-1} 2^i \prod_{j=1}^i s_j. \tag{1}$$

Summing over $\epsilon_0 = \pm 1$ shows that all odd central power sums vanish (this is the complement symmetry $m \mapsto 2^n - 1 - m$ of $T_{n,k}$), and that everything is encoded in the polynomials

$$\widehat{M}_p(n) = \widehat{M}_p(n)(t) := \sum_{s \in \{\pm 1\}^{n-1}} t^{\#\{j: s_j = -1\}} Y(s)^p,$$

via

$$\sum_{m \in T_{n,k}} c(m)^p = \begin{cases} 2[t^k] \widehat{M}_p(n), & p \text{ even,} \\ 0, & p \text{ odd,} \end{cases} \quad S_{n,k,p} = \frac{1}{2^p} \sum_{\substack{j=0 \\ j \text{ even}}}^p \binom{p}{j} (2^n - 1)^{p-j} \cdot 2[t^k] \widehat{M}_j(n). \quad (2)$$

The second identity is the binomial expansion of $m^p = ((c + 2^n - 1)/2)^p$, with the odd- j terms dropped.

2 The triangular recursion and its eigen-expansion

Theorem 1. *The polynomials $\widehat{M}_p$ satisfy, for $n \geq 2$,*

$$\widehat{M}_p(n) = \sum_{j=0}^p \binom{p}{j} 2^j (1 + (-1)^j t) \widehat{M}_j(n-1), \quad \widehat{M}_p(1) = 1. \quad (3)$$

Consequently, since the coefficient matrix is lower triangular with distinct eigenvalues $\lambda_j = \lambda_j(t) = 2^j (1 + (-1)^j t)$,

$$\widehat{M}_p(n) = \sum_{j=0}^p c_{p,j}(t) \lambda_j(t)^{n-1}, \quad (4)$$

where the rational functions $c_{p,j}$ are determined by

$$c_{p,j} = \frac{\sum_{i=j}^{p-1} \binom{p}{i} 2^i (1 + (-1)^i t) c_{i,j}}{\lambda_j - \lambda_p} \quad (j < p), \quad c_{p,p} = 1 - \sum_{j < p} c_{p,j}.$$

Proof. Splitting off the boundary variable s_1 in (1) gives the self-similarity $Y = 1 + 2s_1 \widetilde{Y}$, where $\widetilde{Y}$ is the same functional of $(s_2, \dots, s_{n-1})$. Raise to the p -th power, expand, and sum first over $s_1 \in \{\pm 1\}$ using $\sum_{s=\pm 1} t^{[s=-1]} s^j = 1 + (-1)^j t$, then over the remaining variables; this is (3). The expansion (4) is the standard solution of a triangular linear recursion with distinct eigenvalues; matching the coefficient of λ_j^{n-1} on both sides of (3) gives the formula for $c_{p,j}$ with $j < p$, and the initial condition $\widehat{M}_p(1) = 1$ fixes $c_{p,p}$. $\square$

Theorem 1 answers the algorithmic question at once: iterating (3) computes $\widehat{M}_0, \dots, \widehat{M}_p$ as polynomials in t , hence $S_{n,k,p}$ for every k simultaneously, in $O(np^2)$ arithmetic operations on big integers—versus $\Theta(2^n)$ for enumeration. The expansion (4) is moreover a closed form, uniform in n ; the tables for $p \leq 4$ are as follows (each was generated by exact rational-function arithmetic and verified against exhaustive enumeration; note the $t \mapsto -t$ symmetry between $c_{p,j}$ and $c_{p,p-j}$ up to powers of 2).

$$\widehat{M}_0(n) = (1 + t)^{n-1}, \quad (5)$$

$$\widehat{M}_1(n) = \frac{(1+t)^n - 2^n(1-t)^n}{3t-1}, \quad (6)$$

$$\widehat{M}_2(n) = \frac{(3-t)(1+t)^{n-1}}{3(1-3t)} + \frac{2^{n+1}(1-t)^{n+1}}{(3t-1)(3t+1)} + \frac{4^n(3+t)(1+t)^{n-1}}{3(1+3t)}, \quad (7)$$

$$\widehat{M}_3(n) = \frac{(t-7)(t+1)(1+t)^{n-1}}{(3t-1)(9t-7)} + \frac{(t+3)2^n(1-t)^n}{(3t-1)(3t+1)} + \frac{(t+3)4^n(1+t)^n}{(3t-1)(3t+1)} + \frac{(t-7)8^n(1-t)^n}{(3t-1)(9t-7)}, \quad (8)$$

$$\begin{aligned} \widehat{M}_4(n) = & \frac{(15-t)(7-5t)(1+t)^{n-1}}{15(1-3t)(7-9t)} + \frac{8(7+t)2^{n-1}(1-t)^{n+1}}{(3t-1)(3t+1)(9t+7)} + \frac{8(t^2-9)4^{n-1}(1+t)^{n-1}}{3(9t^2-1)} \\ & + \frac{32(t-7)8^{n-1}(1-t)^{n+1}}{(9t^2-1)(9t-7)} + \frac{16(15+t)(7+5t)16^{n-1}(1+t)^{n-1}}{15(1+3t)(7+9t)}. \end{aligned} \quad (9)$$

Each summand has poles, but the sum is a polynomial in t of degree $n-1$; the singularities cancel identically, which is a useful internal check.

Remark 2 (Krawtchouk form). There is an equivalent “static” formulation that avoids the recursion. Expanding $Y^p = \sum_{i_1, \dots, i_p} 2^{i_1 + \dots + i_p} \prod_j s_j^{e_j}$ with $e_j = \#\{\ell : i_\ell \geq j\}$ and summing over s shows, for even p ,

$$\sum_{m \in T_{n,k}} c(m)^p = 2 \sum_{i_1, \dots, i_p=0}^{n-1} 2^{i_1 + \dots + i_p} K_k(O(i); n-1),$$

where, if $i_{(1)} \leq \dots \leq i_{(p)}$ is the sorted tuple, $O(i) = (i_{(2)} - i_{(1)}) + (i_{(4)} - i_{(3)}) + \dots + (i_{(p)} - i_{(p-1)})$ is the alternating gap sum, and $K_k(d; N) = \sum_j (-1)^j \binom{d}{j} \binom{N-d}{k-j}$ is the binary Krawtchouk polynomial, i.e. $[t^k](1+t)^{N-d}(1-t)^d$. The eigen-expansion results from evaluating this p -fold sum by nested geometric series.

3 Extracting a fixed number of transitions

All denominators in (7)–(9) are products of polynomials *linear* in t , so coefficient extraction is mechanical.

Lemma 3 (extraction). *For any $A, B \geq 0$ and constant a ,*

$$[t^k] \frac{(1+t)^A(1-t)^B}{1-at} = \sum_{i=0}^k a^{k-i} \sum_j (-1)^j \binom{B}{j} \binom{A}{i-j}.$$

Terms with several linear factors in the denominator are first split by partial fractions in t . Applied to (4), this produces, for each fixed even p , an explicit formula for $\sum_{T_{n,k}} c^p$ with $O(k+p)$ terms; the relevant values of a are ± 3 for $p=2$ and $\pm 3, \pm \frac{9}{7}$ for $p=4$.

For $p=2$ the assembled result is compact. Writing $\Sigma_{n,k,2} = \sum_{m \in T_{n,k}} (2m - 2^n + 1)^2$:

$$\begin{aligned} \Sigma_{n,k,2} = 2 \sum_{i=0}^k \left\{ \frac{3^{k-i}}{3} \left[3 \binom{n-1}{i} - \binom{n-1}{i-1} \right] + \frac{(-3)^{k-i} 4^n}{3} \left[3 \binom{n-1}{i} + \binom{n-1}{i-1} \right] \right. \\ \left. - 2^n (-1)^i (3^{k-i} + (-3)^{k-i}) \binom{n+1}{i} \right\}, \end{aligned} \quad (10)$$

and then

$$S_{n,k,2} = \frac{\Sigma_{n,k,2}}{4} + \frac{(2^n - 1)^2}{2} \binom{n-1}{k}. \quad (11)$$

Corollary 4 (low k). *Specializing (10)–(11),*

$$S_{n,1,2} = 4^n \left(n - \frac{7}{3}\right) + 2^{n+1} + n + \frac{1}{3}, \quad S_{n,2,2} = 4^n \left(\frac{n^2}{2} - \frac{17n}{6} + \frac{19}{3}\right) - 2^n (n^2 - n + 10) + \frac{n^2}{2} - \frac{n}{6} + \frac{11}{3}.$$

Remark 5. The value of $S_{n,1,2}$ corrects the formula conjectured in the original question, which omits the “+ n ” term; the two expressions already disagree at $n = 2$, where $T_{2,1} = \{1, 2\}$ gives $S_{2,1,2} = 5$.

Corollary 6 (moments of the uniform distribution on $T_{n,k}$). *Let X be uniform on $T_{n,k}$. Then $\mathbb{E}X = (2^n - 1)/2$, all odd central moments vanish, and*

$$\text{Var}(X) = \frac{[t^k] \widehat{M}_2(n)}{4 \binom{n-1}{k}}, \quad \text{Kurt}(X) = \frac{\mu_4}{\mu_2^2} = \frac{\binom{n-1}{k} [t^k] \widehat{M}_4(n)}{([t^k] \widehat{M}_2(n))^2}.$$

For every fixed k , $\text{Kurt}(X) \rightarrow 1$ as $n \rightarrow \infty$: with few transitions the value is dominated by the top run and the distribution becomes two-lobed. (For $k = 0$ and $k = n - 1$ the kurtosis is exactly 1.) Sample values at $n = 64$: 1.0297 at $k = 1$, 1.2302 at $k = 8$, 1.7869 at $k = 31$.

Remark 7 (on “closed form”). For each fixed p , (4) with Lemma 3 is an exact formula with $O(k+p)$ terms, uniform in n and k . A single hypergeometric-free expression in all three parameters simultaneously seems out of reach: the k -slices irreducibly involve the partial sums $\sum_{i \leq k} 3^i \binom{n}{i}$ (and $\frac{9}{7}$ -weighted analogues at $p = 4$), which do not telescope.

4 Striped generators: independent transitions

The generators that motivated the question do not fix k . A *striped* generator with mean run length $m \geq 2$ draws b_0 uniformly and then repeats the previous bit with probability $1 - \theta$, where $\theta = 1/m$; that is, the s_j of Section 1 are i.i.d. with $\Pr[s_j = -1] = \theta$. Set

$$\beta = \mathbb{E}[s_j] = 1 - 2\theta = 1 - \frac{2}{m}.$$

The analysis of Theorem 1 goes through verbatim with the marker t replaced by expectation: the recursion for $\mathbb{E}[Y^p]$ has the triangular matrix $\binom{p}{j} 2^j e_j$ with $e_j = 1$ for even j and $e_j = \beta$ for odd j , hence eigenvalues $1, 2\beta, 4, 8\beta, 16, \dots$. For the centered value $c = 2X - (2^n - 1)$ all odd moments vanish, and the even ones are short rational expressions:

$$\mathbb{E}[c^2] = \frac{1 + 2\beta}{3(2\beta - 1)} + \frac{4\beta^2 (2\beta)^{n-1}}{(2\beta - 1)(\beta - 2)} + \frac{4(2 + \beta) 4^{n-1}}{3(2 - \beta)}, \quad (12)$$

$$\begin{aligned} \mathbb{E}[c^4] = & \frac{(6\beta + 1)(8\beta + 7)}{15(2\beta - 1)(8\beta - 1)} - \frac{8\beta^2(3\beta + 4) (2\beta)^{n-1}}{(2\beta - 1)(2 - \beta)(8 - \beta)} - \frac{8(2\beta + 1)(\beta + 2) 4^{n-1}}{3(2\beta - 1)(\beta - 2)} \\ & + \frac{32\beta^2(4\beta + 3) (8\beta)^{n-1}}{(\beta - 2)(8\beta - 1)(2\beta - 1)} + \frac{16(7\beta + 8)(\beta + 6) 16^{n-1}}{15(8 - \beta)(2 - \beta)}, \end{aligned} \quad (13)$$

with $\text{Var}(X) = \frac{1}{4} \mathbb{E}[c^2]$ and $\text{Kurt}(X) = \mathbb{E}[c^4] / \mathbb{E}[c^2]^2$.

Remark 8 (removable singularities). The parameter values $\beta = \frac{1}{2}$ (mean run length $m = 4$) and $\beta = \frac{1}{8}$ are removable singularities of (12)–(13), caused by eigenvalue collisions ($2\beta = 1$, resp. $8\beta = 1$); evaluate by taking limits, or by running the recursion directly. For example, at $m = 4$,

$$\mathbb{E}[c^2] = \frac{5 \cdot 4^n - 6n - 5}{9}.$$

Corollary 9 (limits). *As $n \rightarrow \infty$,*

$$\text{Var}\left(\frac{X}{2^n}\right) \rightarrow \frac{2 + \beta}{12(2 - \beta)}, \quad \text{Kurt}(X) \rightarrow \frac{3(2 - \beta)(6 + \beta)(8 + 7\beta)}{5(8 - \beta)(2 + \beta)^2}.$$

At $\beta = 0$ (unstriped, uniform bits) these are $\frac{1}{12}$ and $\frac{9}{5}$, the variance and kurtosis of the uniform distribution; as $\beta \rightarrow 1$ they tend to $\frac{1}{4}$ and 1, those of a symmetric two-point distribution.

m	β	$\lim \text{Var}(X/2^n)$	$\lim \text{Kurt}(X)$	
2	0	$1/12 \approx 0.0833$	$9/5 = 1.8$	(uniform)
3	$1/3$	$7/60 \approx 0.1167$	$1767/1127 \approx 1.5679$	
4	$1/2$	$5/36 \approx 0.1389$	$897/625 = 1.4352$	
8	$3/4$	$11/60 \approx 0.1833$	≈ 1.2234	
16	$7/8$	$23/108 \approx 0.2130$	≈ 1.1130	
32	$15/16$	$47/204 \approx 0.2304$	≈ 1.0568	

5 Verification

All results were checked in exact rational arithmetic: the eigen-expansions (7)–(9) against exhaustive enumeration of all 2^{n-1} transition strings for $p \leq 4$ and $n \leq 12$; the bridge (2) against the raw integer sums $\sum_m m^p$ over $[0, 2^n)$ for $n \leq 10$ and all k ; the closed forms against an independent iteration of (3) at $n = 40$; the explicit formulas (10)–(11) and Corollaries 4–6 against brute force on the stated ranges; and (12)–(13) against probability-weighted enumeration at $\beta \in \{\frac{1}{3}, \frac{3}{5}, \frac{15}{16}, -\frac{1}{3}\}$.